

UET University Peshawar

Abbottabad Campus



Lab Report No: 08

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Course: SE-305L Computer Communications and Networks

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Date: 18 June 2026

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Lab 08: Designing a Network for a Smart Home Environment

Objective:

To create a basic smart home network using Cisco Packet Tracer IoT devices, register all devices to a wireless Home Gateway, and manually control IoT devices to simulate smart home behaviour.

- Lab Tasks:
1. Create rooms using labels or shapes
 2. Add smart (IoT) devices like lights, doors, cameras
 3. Add a wireless home gateway (router)
 4. Register each IoT device to the home gateway
 5. Add a smart phone or tablet for control
 6. Verify Device Registration and Connectivity
 7. Manual Control of IoT Devices via Tablet/Smartphone
 8. Experiment with Device Placement and Connectivity
 9. Report

Required Devices:

- Home Gateway
- Smart Light
- Motion Sensor
- Smart Door
- Smart Thermostat
- Tablet or Smartphone

Instructions

Step 1: Create Room Labels or Zones

You can organize the smart home visually:

- Use the 'Place Note' tool (top right, sticky note icon, or press N) to label:
 - o Living Room(s)
 - o Bedroom(s)

o Kitchen(s)

Step 2: Add a Home Gateway (Router + Wi-Fi)

- Network Devices > Wireless Devices or Ctrl+Alt+W
- Drag a device called "Home Gateway" into your workspace.

This will provide internet and wireless connectivity to all smart devices.

Step 3: Add IoT Devices (Smart Devices)

Go to the End Devices > Home or Ctrl+Alt+H and drag in any of the following:

- Motion Sensor
- Smart Light
- Smart Door
- Smart Thermostat
- Camera
- Fan or TV Place them near their respective rooms.

Step 4: Add a Smart Phone or Tablet

- Go to "End Devices"
- Add a Tablet or Smartphone to control your smart home.

Step 5: Register IoT Devices to Home Gateway

Each IoT device must be registered to the Home Gateway for it to work. Here's how:

Example: Register the Smart Light

1. Click on the Smart Light
2. Go to the "Config" tab
3. Under "Wireless0": Make sure the SSID is set to HomeGateway (this is usually the default).
4. Ensure the "On" checkbox is selected.
5. If everything is correct, the device will automatically connect to the Home Gateway.

Repeat this for each smart device and your Tablet or Smart Phone.

Step 6: Verify Device Registration and Connectivity

For each IoT device (Smart Light, Thermostat, Door, etc.),

1. Click on the device → Go to the Config tab → Under IoT Server, select Home Gateway.
2. The Connected button may remain disabled, but this is normal. The device connects automatically in the background.
3. To confirm connection, open the Tablet or Smartphone device → Launch IoT Monitor → Check if the devices appear in the device list.

Step 7: Manual Control of IoT Devices via Tablet/Smartphone

1. On the Tablet or Smartphone, select a device from the list (e.g., Smart Light).
2. Use the app interface to turn the device ON or OFF, or adjust settings such as:
 - o Thermostat temperature
 - o Door lock status
3. This simulates manual control of the smart home devices through the Home Gateway.

Step 8: Experiment with Device Placement and Connectivity

1. Try moving IoT devices farther from the Home Gateway to observe connectivity changes.
2. Add additional IoT devices like fans, cameras, or smart TVs and connect them to the Home Gateway.
3. Explore different room placements to see how device management appears on the tablet.

Step 9: Report Write a short report including:

1. The list of devices used and their placement.
 - *Motion Sensor*
 - *Smart Light*
 - *Smart Door*
 - *Smart Thermostat*
 - *Camera*
 - *Fan or TV*
2. Confirmation of device connectivity via the tablet/smartphone.

Steps to Confirm Connectivity via Smartphone/Tablet (SMARTPHONE-PT)

1. **Click on the Smartphone device (Smartphone0)** in your Packet Tracer simulation.
2. Go to the **“Desktop”** tab.
3. Open the **“IoT Monitor”** application.
4. You should see a list of connected IoT devices (like:
 - o Light (IoT9),
 - o Ceiling Fan,
 - o Thermostat (IoT6),
 - o Motion Detector (IoT5),

- Webcam (IoT7),
- Door,
etc.).
- 5. **Verify connectivity:**
 - Each device listed should show a status such as **Online** or **Connected**.
 - If a device is **not connected**, it may not appear or might show **Offline**.
- 6. You can also go to the **"IoT Service" tab** to manually control devices to confirm they are responsive (e.g., toggle the light on/off, or adjust thermostat).

3. How manual control was tested and which devices you controlled.

✓ Manual Control Testing Steps:

1. **Open Smartphone0:**
 - Click on the smartphone device in your workspace.
 - Go to the **Desktop** tab.
 - Launch **IoT Monitor** or **IoT Custom GUI**.
2. **Connect to Home Gateway** (if not auto-connected):
 - In the IoT Monitor, click "Connect" or ensure it's connected to Home Gateway0.
3. **List of Connected Devices Appears:**
 - You will see a list of all connected IoT devices, such as:
 - **IoT5: Motion Detector**
 - **IoT6: Thermostat**
 - **IoT7: Webcam**
 - **IoT9: Light**
 - **Fan**
 - **Door**
4. **Manual Control Test:**

For each listed device, tap or select it and perform these actions:

 - **Light (IoT9):**
 - Click "On" → the light should turn on.
 - Click "Off" → the light should turn off.
 - **Ceiling Fan:**
 - Set speed or toggle power.

- Observe change (e.g., moving blades in simulation).
 - **Thermostat (IoT6):**
 - Change mode (Cool/Heat) or temperature.
 - Check if values are updated.
 - **Door:**
 - Open/Close the door manually.
 - **Webcam (IoT7):**
 - Enable or check motion streaming status (if supported).
 - **Motion Detector (IoT5):**
 - Verify its status (e.g., active/inactive) via the monitor interface.
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4. Screenshots.

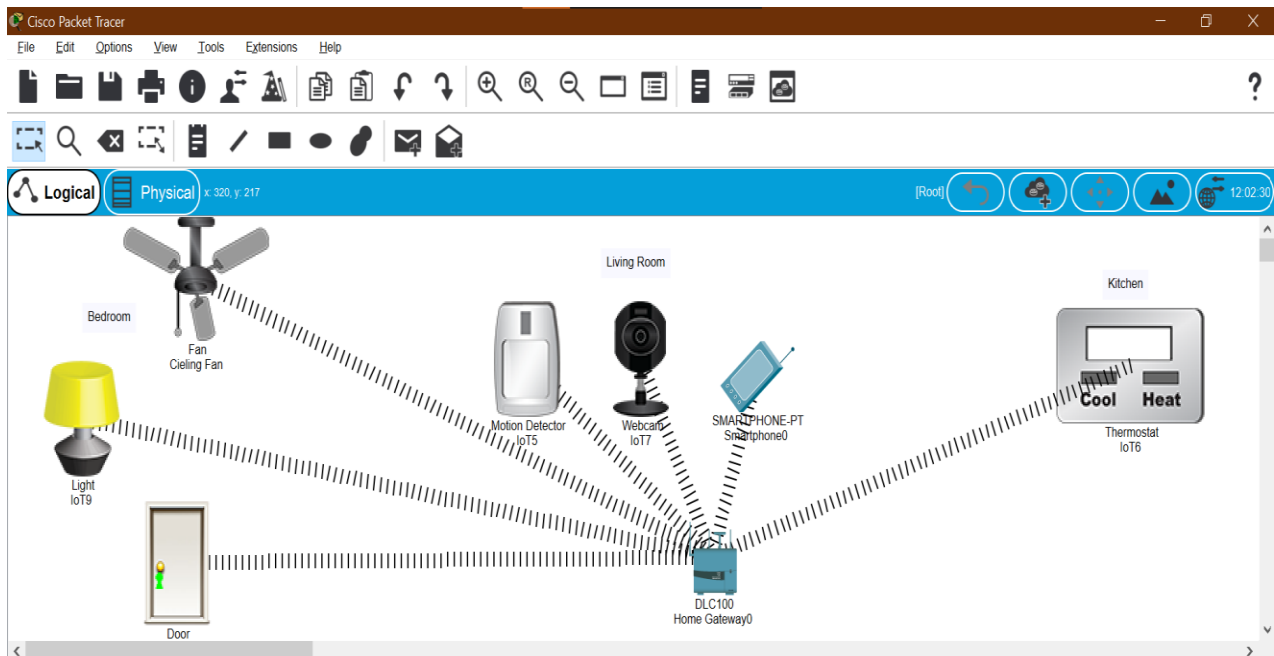


Figure 1 Network Simulations

SpecificationsI/O ConfigPhysicalConfigThing EditorProgrammingAttributes

Network Adapter

PT-IOT-NM-1W

Network Adapter 2

None

Digital Slots

1

Analog Slots

0

USB Ports

0

Bluetooth

☐ Built-in

Desktop

☐ Show

Usage

☒ Smart Device☐ Component

SpecificationsI/O ConfigPhysicalConfigThing EditorProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Global Settings

Display Name

IoT6

Serial Number

PTT0810D2F0-

Gateway/DNS IPv4

☒ DHCP☐ Static

Gateway

192.168.25.1

DNS Server

0.0.0.0

Gateway/DNS IPv6

☒ DHCP☐ Auto Config☐ Static

IPv6 Gateway

IPv6 DNS Server

IoT Server

☐ Top

Advanced